with respect to their priority: in figure 3.2 you can see that children nodes always have a higher stock count than their parents, but there is no ordering between siblings.

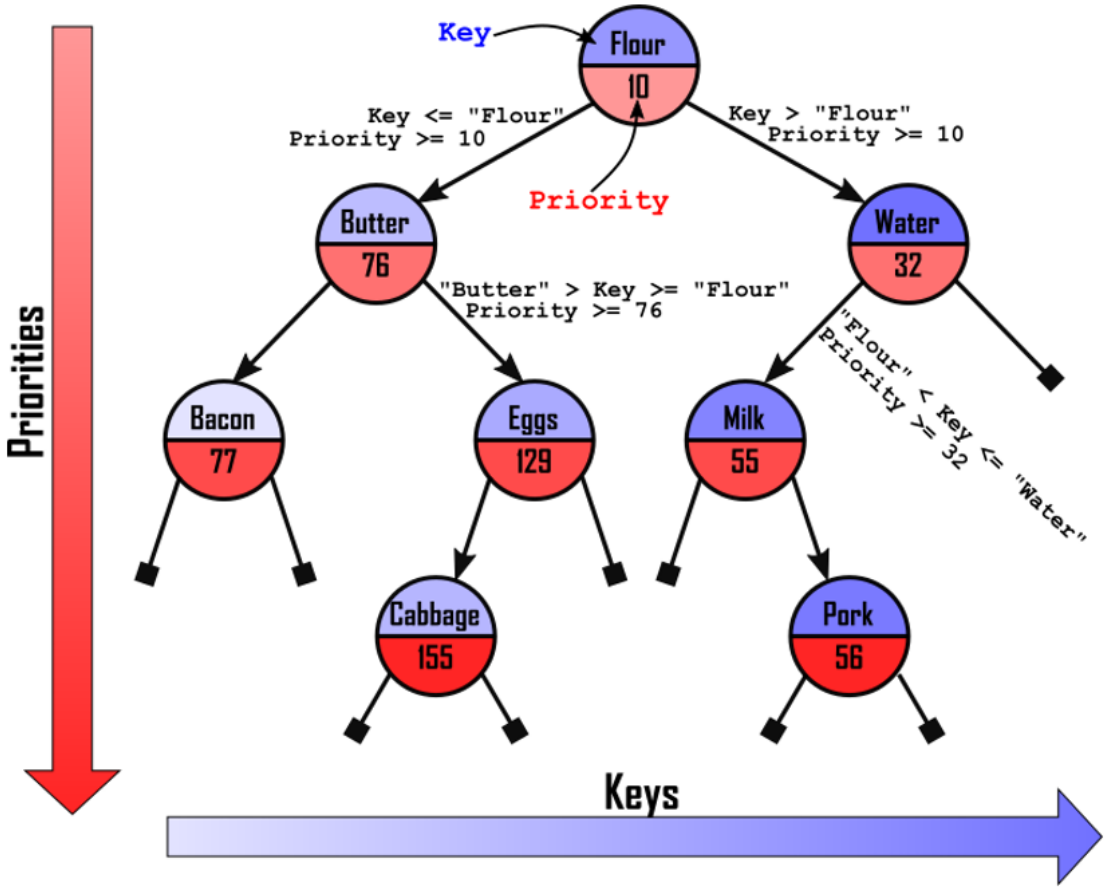


Figure 3.2 An example of a treap, with strings as BST keys and integers as heap priorities. Note that the heap, in this case, is a min-heap, and so smaller numbers go on top.

This kind of tree offers an easy way to query entries by key (by the products names, in the example), while we couldn't easily run a query on priorities; we can, however, efficiently locate the element with the highest priority (the smallest inventory count): it will always be at the root of the tree!

Extracting the top element however… it's going to be more complicated than with heaps! We can't just replace it with a